Dimple Sarnaaik

Department of Physics & Astronomy, University of Southern California
Los Angeles, CA 90007, USA
Carnegie Observatories, Pasadena, CA, USA
sarnaaik@usc.edu

Research Interests: **Astrophysical probes of dark matter**. Near-field cosmology and the subhalo mass function. Beyond-CDM dark matter models. Calibrating beyond-CDM (WDM, IDM, FDM) merger trees against N-body simulations for use in semi-analytic models (Galacticus). Constraining dark matter substructure with Roman Space Telescope stellar astrometry. Dark matter–baryon interactions from CMB and large-scale structure. Statistical inference and machine learning for dark matter model selection.

Education

2021– **University of Southern California**, Los Angeles, CA
Ph.D. in Physics
Expected Graduation: Spring 2027
- Thesis: *Probing the Dark Universe: A Multi-Scale Framework for Beyond-CDM Constraints*
- Advisors: Andrew Benson (Carnegie Observatories) & Kris Pardo (USC)

2021 **University of Southern California**, Los Angeles, CA
B.S. in Physics & Astronomy
- Cumulative GPA: 3.76/4.00 | Astrophysics GPA: 3.9/4.00

Research Experience

Mocks of EDGE Dwarf Galaxies for Roman-era SBF Tests
Doctoral Researcher (with Stacy Kim; EDGE collaboration)
- Early-stage work: building mock images of dwarf galaxies from the EDGE hydrodynamical simulations for the Nancy Grace Roman Space Telescope

Beyond-CDM Halo Mass Functions and Merger Trees
Doctoral Researcher (Advisor: Andrew Benson)
- Calibrating beyond-CDM (WDM, IDM, FDM) progenitor mass functions against the COZMIC N-body suites
- Testing effect of tophat and sharp-k filter transfer functions on merger tree parameters
- Incorporated an improved backsplash halo identification algorithm for subhalo finding in N-Body simulations

Dark Matter Subhalo Detection with Roman Astrometry
Doctoral Researcher (Advisor: Kris Pardo)
- Built a forward model pipeline for the Nancy Grace Roman Space Telescope to detect dark matter subhalos via gravitational perturbations of stellar apparent accelerations
- Forecasting Roman's sensitivity to the subhalo mass spectrum for CDM, WDM, and PBH models with improved astrometric precision with longer observation times

Dark Matter–Electron Scattering Constraints from CMB and Large-Scale Structure
Undergraduate Researcher (Advisor: Vera Gluscevic)
- Derived competitive upper limits on dark matter–electron scattering cross-sections using CMB and large-scale structure data, placing new bounds on interacting dark matter models
- Co-first author: *Nguyen, Sarnaaik et al.* 2021, Phys. Rev. D

Publications

3 total (1 refereed, 2 first-author manuscripts in preparation). Science-writing articles noted separately.

Refereed

Nguyen, Q.[†], **Sarnaaik, D.**[†], Boddy, K. K., Nadler, E. O., Gluscevic, V. *"Observational constraints on dark matter scattering with electrons"*, 2021, Phys. Rev. D, 104, 103521

[†] Co-first author.

In Preparation

Sarnaaik, D., Pardo, K. *"Probing Dark Matter Subhalo Populations with the Nancy Grace Roman Telescope"*, in prep

Sarnaaik, D., Benson, A. *"Beyond-CDM Merger Tree Calibration with Galacticus"*, in prep

Science Writing

Sarnaaik, D. "POV: The life of an academic." *The Lab Report*, Vol. 3, 2026. USC Department of Physics & Astronomy. [link]

Sarnaaik, D. "Astronomy across barriers." *The Lab Report*, Vol. 2, 2025. USC Department of Physics & Astronomy. [link]

Presentations

Invited Talks

2027	UCLA Lunch Talk	Los Angeles, CA
2026	KIPAC Tea Talk, Kavli Institute for Particle Astrophysics & Cosmology	Stanford University, CA
	Pasadena City College — Guest Lecture	Pasadena, CA
	KJ Somaiya School	Mumbai, India
	Mandsaur High School	Mandsaur, India

Contributed Talks

2026	Galaxy Formation & Evolution in Southern California (GalFRESCA)	UC Irvine, CA
	Dark Matter Probe Combination Workshop, KICP	Chicago, IL
2025	APS April Meeting	Anaheim, CA

Posters

2026	American Astronomical Society (AAS) Meeting — **Chambliss Award**	Passadena, CA
2025	UCLA Dark Matter Conference	Los Angeles, CA
2024	Conference for Undergraduate Women in Physics (CUWiP)	San Diego, CA
	oSTEM Annual Conference	Anaheim, CA
2023	USC Physics Department Research Conference	Catalina Island, CA
	Galaxy Formation & Evolution in Southern California (GalFRESCA)	Riverside, CA

Honors & Awards

Honors

2026	Chambliss Astronomy Achievement Student Award, American Astronomical Society	—
2025	Rockwell Dennis Hunt Scholastic Award, USC	$5,000
2025	Wall of Scholars (Rockwell Dennis Hunt Scholastic Award 2025), USC	—
2024	Galactic Leader Award, Cosmolab	—
2023	Rockwell Dennis Hunt Scholastic Award, Honorable Mention, USC	$1,000

Funding

2026	Travel Fellowship, USC Physics & Astronomy Department	$750
2023–24	USC Dornsife Graduate Fellowship, USC Dornsife College	$35,700
2022–24	GSG Professional Development Fund (Spring 2022, Spring 2023, Fall 2023, Fall 2024), USC	$500 each
2020–21	Goel Family Scholarship (USC Dornsife Continuing Student Scholarship Award)	$5,000
2020	Summer Undergraduate Research Fund (SURF), USC	$3,000
2020	Student Opportunities for Academic Research (SOAR), USC	$1,000
2019	Student Opportunities for Academic Research (SOAR), USC	$500
2018–20	Women in Science & Engineering Research Fellowship (2018, 2019, 2020), USC	$500 each

Teaching

2021– **Teaching Assistant**, University of Southern California
- 8 courses across 5 years; 4–6 lab sections per semester (∼20 students each), ranging from introductory physics to upper-division astronomy

2019–21 **Resident Assistant**, University of Southern California
- Managed programming and student support for a residential community of ∼120 undergraduates

2019–20 **Peer Tutor**, USC Athletic Academic Services
- Provided one-on-one STEM tutoring to ∼15 student-athletes in physics, math, and astronomy

Professional Service & Leadership

Departmental & Student Organization Service

2025 **Judge**, USC Undergraduate Research Symposium, USC
2024– **Co-Founder**, Women+ in Physics (W+iP), USC
2022–24 **President**, Graduate Astronomy Student Peers (GASP), USC
2019–21 **Co-President**, Society of Physics Students (SPS), USC

Outreach

2026 **Foster Elementary School Partnership**, Los Angeles, CA
- In-person astronomy demonstrations for elementary school students in South Los Angeles

2025– **Science Journalist**, *The Lab Report*, USC Physics & Astronomy
- "POV: The life of an academic" (Vol. 3, 2026) and "Astronomy across barriers" (Vol. 2, 2025); ongoing contributor to USC's departmental science journalism platform

2023–24 **Skype a Scientist**
- Visited 7 K–8 classrooms virtually; topics included dark matter, black holes, phases of the moon, and Earth's seasons

2023– **Letters to a Pre-Scientist**
- Pen-pal with middle school students on what a career in physics research actually looks like